

Environmental fluoxetine promotes skin cell proliferation and wound healing.

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Abstract Pharmaceutical pollutants, including the antidepressant fluoxetine (FLX, commercial name: Prozac), are increasingly detected in aquatic environments, raising concerns about their effects on non-target organisms, including humans. The original study demonstrated that environmentally-relevant FLX concentrations promote wound closure in human skin biopsies and keratinocytes through enhanced cell proliferation and serotonin signalling, supported by transcriptomic, phosphoproteomic, and kinase profiling analyses. Building on these findings, the present study extends the transcriptomic analysis by examining gene expression in scratched HaCaT keratinocytes exposed to 540 ng/L FLX at two time points: 6 and 36 hours. Differentially expressed genes (DEGs) were identified at each time point and compared to unexposed control cells, with an intersection analysis performed between both conditions. We identified 147 DEGs at 6 hours and 250 DEGs at 36 hours, with upregulated genes associated with cell proliferation and cellular stress defence mechanisms. The difference in DEG numbers compared to the original study reflects sample quality filtering applied during data processing. These results provide a more detailed temporal resolution of FLX-induced transcriptomic changes and support the conclusion that environmental FLX exposure modulates gene expression in ways that may meaningfully impact skin wound healing.

Introduction

Pharmaceutical pollution has become a growing environmental concern, particularly the presence of antidepressants in aquatic ecosystems. Fluoxetine (FLX), commercially known as Prozac, is one of the most widely prescribed selective serotonin reuptake inhibitors (SSRIs) and works by blocking the reabsorption of serotonin in the brain, thereby increasing serotonin availability and improving mood regulation. Due to incomplete metabolism in the human body and inadequate removal by wastewater treatment plants, FLX is consistently detected in surface waters at concentrations ranging from nanograms to micrograms per litre [R Core Team \(2012\)](#)(Metcalf et al., 2010).

Serotonin is not only a neurotransmitter but also plays a well-established role in peripheral tissues, including the skin, where it is involved in regulating cell proliferation and wound healing. Wound healing is a complex and tightly coordinated biological process that proceeds through three overlapping phases: inflammation, proliferation, and tissue remodelling. During the proliferation phase, keratinocytes — the dominant cell type of the outermost skin layer (the epidermis) — migrate toward the wound edge and actively divide to restore the skin barrier. Any disruption or enhancement of this process can have significant clinical implications. Given that FLX modulates serotonin signalling, and serotonin itself plays a role in keratinocyte activity, it is biologically plausible that environmental FLX exposure could meaningfully influence skin repair mechanisms.

The original study by [\(Rodriguez-Barucg et al., 2024\)](#) was the first to investigate this link experimentally, using a multi-omics approach combining transcriptomics, phosphoproteomics, and kinase profiling. Their transcriptomic analysis identified 350 differentially expressed genes (DEGs) in FLX-exposed keratinocytes, with upregulated genes linked to cell proliferation and tissue morphogenesis, and downregulated genes associated with mitochondrial function and metabolism [\(Rodriguez-Barucg et al., 2024\)](#). These findings provided a molecular basis for the observed enhancement in wound closure, but left room for deeper temporal investigation of how gene expression changes evolve over time during FLX exposure.

Transcriptomics is the large-scale study of all RNA molecules expressed in a cell or tissue at a given moment, essentially providing a snapshot of which genes are active and to what degree. By comparing gene expression between exposed and unexposed cells, it is possible to identify which biological processes are being switched on or off in response to a treatment. In this study, we use RNA sequencing (RNA-seq) data analysed with DESeq2, a statistical tool specifically designed to detect differentially expressed genes from count-based sequencing data while accounting for biological variability between samples.

The present study focuses exclusively on the transcriptomic component of the original research. We analysed gene expression in scratched HaCaT human keratinocytes exposed to 540 ng/L FLX at two time points: 6 and 36 hours, mirroring the time points used in the transcriptomic section of the original study [\(Rodriguez-Barucg et al., 2024\)](#). The 6-hour time point captures the early transcriptional response to FLX exposure, reflecting immediate cellular adjustments, while the 36-hour time point represents a later stage where more sustained gene expression changes associated with active wound closure are expected. Our central research question is: does FLX exposure significantly alter gene

expression in human keratinocytes in ways that support enhanced wound healing, and how does this response evolve between 6 and 36 hours of exposure? To address this, DEGs were identified at each time point and compared to unexposed control cells, followed by enrichment analysis to map these genes to relevant biological pathways. An intersection analysis was further performed to identify genes consistently differentially expressed across both time points. We also explored whether enrichment analysis could reveal additional gene functions and pathways beyond those reported in the original article (Rodriguez-Barucg et al., 2024).

Materials and Methods

Cells and clinical samples

HaCaT cells are immortalised human keratinocytes derived from normal adult human skin and were obtained from AddexBio. Due to their well-characterised behaviour and relevance to skin biology, HaCaT cells are widely established as a model system in dermatological research. Cells were maintained in 5% CO₂ incubation chambers at 37 °C and subcultured every 3 days or upon reaching approximately 80% confluency.

To investigate the transcriptomic effects of Fluoxetine (FLX) on wound healing, a scratch assay was performed on HaCaT cell monolayers. Following the scratch, cells were treated with FLX at a concentration of 540 ng/L. Conditions were defined as a combination of treatment (FLX vs. control) and time post-scratch (6 h and 36 h), resulting in a total of four samples across both conditions.

RNA sequencing

HaCaT cells were scratched three times per well in 6-well plates, and total RNA was isolated at 6 h and 36 h post-scratch using the PureLink™ RNA Mini Kit (Thermo Fisher Scientific), following the manufacturer's instructions. Library preparation and RNA sequencing were carried out by Novogene Co. Ltd. (Cambridge, UK) using the Illumina NovaSeq 6000 platform. The sequencing output consisted of two files, one for each comparison (6 h and 36 h), each containing one FLX-exposed sample and one untreated control sample. Each file included gene annotation columns (gene name, gene ID, chromosome, start and end positions, strand, length, and biotype), raw read counts per sample, FPKM-normalised expression values, and statistical outputs including log₂log₂ fold change, p-value, and adjusted p-value (padj). Raw read counts were used as input for all downstream differential expression analysis.

Most likely this will be the largest section in your report (depending on the number of images you show in the Results section). Take a look at your experiments article to see how they organised its content to see if this is also suitable for your report.

1. This section must at the very least contain information about the **data**; how many samples from what organism + strain (if applicable), what kind of data is there available (not only saying FPKM but also the sequencer type that generated this data).
2. Also, information about all the **software** that you used during your analysis (i.e. R and its libraries, bioConductor, DESeq2, etc.) and all **methods** that you applied (both filtering steps and statistical analysis steps). Try to use the proper terms and symbols if needed, i.e. *"We filtered for DEGs with an $\alpha < 0.05$ "*. **Symbols** need to be surrounded with \$ signs, otherwise you will get an error when knitting.

Do not show any results in this section, for instance when stating that you filtered t-test p-values on a certain value, you do not list how many genes were left, that is for the next section.

Data Pre-processing

Prior to differential expression analysis, sample quality was assessed using multiple visualizations. Raw count data were normalized using Variance Stabilizing Transformation (VST) as implemented in DESeq2, and Poisson distance-based normalization was applied to measure dissimilarity between samples. Sample-to-sample distances were visualized using heatmaps and Multidimensional Scaling (MDS) plots, generated with PoiClu, ggplot2 and pheatmap in R, alongside DESeq2 built-in plotting functions. Additionally, bar plots and density plots were produced to inspect the distribution of counts across samples.

Based on these quality control visualizations, three irregular samples were identified and removed from the analysis: C36-R1 and E36-R3 from the 36-hour dataset, and C6-R4 from the 6-hour dataset.

Sample removal was performed while ensuring that each experimental group retained a minimum of three biological replicates, in order to maintain statistical reliability in the downstream analysis.

Differential Expression & Enrichment Analysis

Differential gene expression analysis was performed separately for each time point (6 h and 36 h) using the DESeq2 R package. Genes were considered differentially expressed at an unadjusted p-value threshold of $p < 0.05$. It should be noted that unadjusted p-values were used in place of adjusted p-values (padj), as this approach yielded results most comparable to those reported in the original study.

Entrez gene IDs (ENTREZID) were used as the primary gene identifier for all enrichment analyses, as they provide a standardised identifier supported across biological databases. Three complementary enrichment analyses were performed using the `clusterProfiler` and `ReactomePA` R packages, with gene annotations provided by `org.Hs.eg.db`, all applying a significance threshold of $p < 0.05$ without p-value adjustment:

- **Gene Ontology (GO) enrichment (enrichGO)** was performed across all three ontology domains — Biological Process (BP), Cellular Component (CC), and Molecular Function (MF) — as each domain captures a distinct and complementary aspect of gene function.
- **KEGG pathway enrichment (enrichKEGG)** was applied to identify which known biological pathways were significantly represented among the DEGs, with gene set sizes restricted between 10 and 500 genes.
- **Reactome pathway enrichment (enrichPathway)** was used to identify precise molecular reactions and sub-pathways driven by the DEGs.

Results

Here you will only focus on your own results, start at the beginning but only show the interesting results. This section works best if you alternate between text introducing some result, a figure or table showing the result, and a short text segment analyzing the presented result *and* introducing the next figure, etc.

Although we have worked in R a lot, the results are almost never pieces of R code. If you have written your own algorithm or analysis method try to present it in a formula *followed* by the R implementation, otherwise this section should mainly contain figures and tables.

There are multiple methods of including a figure in this section. Figure 1 shows the `LaTeX` method (see the source code) giving very fine control over size and placement of a figure. The size of a figure is determined by the width of the text column, Figure 1 is scaled as 85% of the text width. The [Wikibooks.org](https://www.wikibooks.org/wiki/LaTeX/Tables) website shows some examples for placement (for instance, what the `[htbp]` means), scaling and captions.

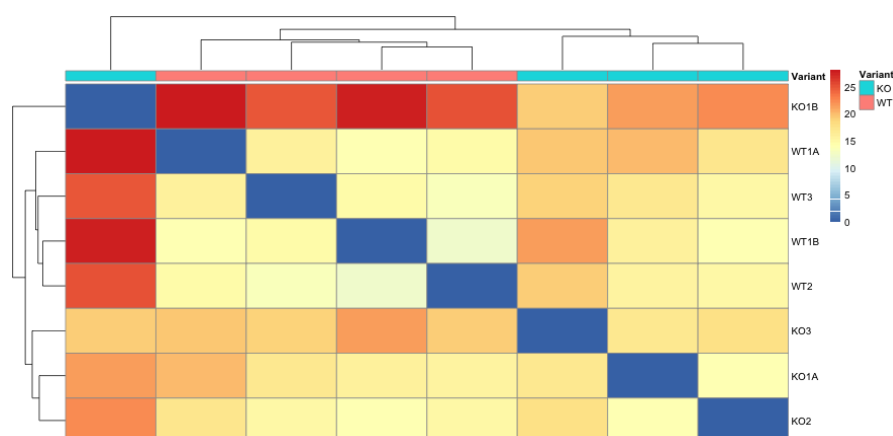
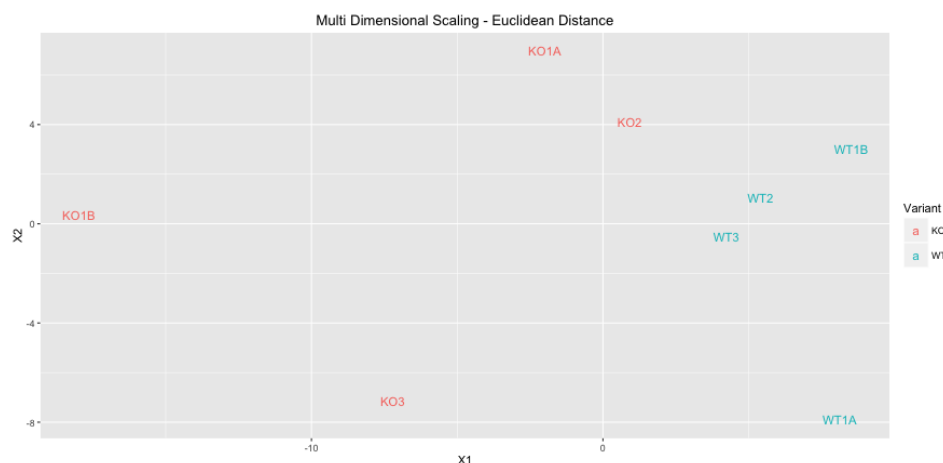


Figure 1: Heatmap showing sample distances and clustering.

Alternatively the previously shown R method using the `PNG` library can be used which allows changing the *height* and *width* of an image but lacks the possibility of referencing the figure in the text, aligning the figure (the heatmap is centered), providing a figure caption and automatic numbering of figures.

```
library(png)
library(grid)
img <- readPNG("images/mds.png")
grid.raster(img)
```



Conclusion

Limit to your *own* conclusions, based on what is shown in the **Results** section. Do not include any comparison to the original article as that is the content of the next section, the **Discussion**. Usually you refer back to all steps outlined in the results section since you included it there for a reason.

Discussion

An important section in your report; here you can compare your findings with the findings presented in the original article. If you found exactly the same results - and that was your goal from the start - this will be a bit shorter section and you should focus on other parts of your research that hasn't been discussed or shown in the article, i.e. any findings from the exploratory data analysis or differences in DEGs using the manual- and library-method.

Besides comparing the two 'experiments', this is also the place to discuss how reliable your results are. Did anything go wrong or did you get unexpected outcomes using some method and if so, what did you do to solve it or are there still open discussion points on this subject.

Notes on the bibliography section

The last section in the report contains all the used references that are used throughout the text in a section called the `Bibliography`. In the text you can link to references using the `LATEX` commands (see the source of this file for instructions). These references must be present in a file named `RJreferences.bib` which is automatically generated once you click the `Knit` button for the first time in a folder with the same name as the current markdown file (`./report-template/RJreferences.bib` for this example file).

This file containing the references can be edited in RStudio too, browse to the folder (after Knitting) and click on the `.bib` file to edit. If, for example, I want to add the article that is available for the experiment used in the week 4 - 7 assignment document, I would first lookup the paper in [PubMed](#) to get the details and add these details to the `.bib` file as follows:

```
@article{garcia,
  author = {Duart-Garcia, Braunschweig},
  title = {Functional expression study of igf2 antisense transcript in mouse},
  journal = {International Journal of Genomics},
  doi = {10.1155/2014/390296},
```

```

url = {http://dx.doi.org/10.1155/2014/390296},
year = 2014
}

@article{rodriguez,
  author = {Quentin Rodriguez-Barucg, Angel A. Garcia, Belen Garcia-Merino, Tomilayo Akinmola, E. Bringas, I. Ortiz, M. A. Wade, A. Dowle, D. A. Joyce, M. J. Hardman, H. N. Wilkinson, and P. Beltran-Alvarez},
  title = {Environmental fluoxetine promotes skin cell proliferation and wound healing},
  journal = {Environmental Pollution},
  doi = {10.1016/2024/124952},
  url = {https://doi.org/10.1016/j.envpol.2024.124952},
  year = 2024
}

```

References added to this file do not automatically show up in your document, unless you actually refer to it. To add a reference, like so: (Duart-Garcia, 2014), you need to use the `\citep{}` or `\citer{}` $\LaTeX$ command combined with the given ID which is `garcia` in our case.

Bibliography

- B. Duart-Garcia. Functional expression study of igf2 antisense transcript in mouse. *International Journal of Genomics*, 2014. doi: 10.1155/2014/390296. URL <http://dx.doi.org/10.1155/2014/390296>. [p5]
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- Q. Rodriguez-Barucg, A. A. Garcia, B. Garcia-Merino, T. Akinmola, T. Okotie-Eboh, T. Francis, E. Bringas, I. Ortiz, M. A. Wade, A. Dowle, D. A. Joyce, M. J. Hardman, H. N. Wilkinson, and P. Beltran-Alvarez. Environmental fluoxetine promotes skin cell proliferation and wound healing. *Environmental Pollution*, 2024. doi: 10.1016/2024/124952. URL <https://doi.org/10.1016/j.envpol.2024.124952>. [p1, 2]

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